



Q-TRACE: ARCHITECTURE & USER FLOWS

Technical Blueprint & Pedagogical Sequence Specifications for SIH26140

NATIONAL QUANTUM MISSION

MINISTRY OF EDUCATION (AICTE)

PRODUCTION CERTIFIED

1. System Architecture

Multi-engine simulation, AST security sandbox, and evidence-bound AI.

2. Safety & Trust Gateway

Zero-hallucination validation and arbitrary code injection rejection.

3. Core Learner Loop

Autonomous Predict → Simulate → Inspect → Repair pedagogical journey.

4. Institutional Governance

Cohort telemetry and 1-click faculty remedial intervention dashboard.

ARCHITECTURE FULL-STACK BLUEPRINT

TIER 1: CLIENT APPLICATION LAYER

Next.js 15+ • App Router • React 19 • TypeScript Strict • Tailwind v4 • TanStack Query

- Learner App Shell**
Gamified serpentine ChamberNodes, responsive 12-col layout, and theme engine.
- Circuit Workspace**
@dnd-kit drag-and-drop grid with bidirectional CodeMirror 6 Python sync.
- Live Quantum Linter**
Instant checks: Q1-1 (post-collapse), Q1-2 (no-cloning), Q1-3 (wire collisions).
- Visual Evidence Studio**
Plotly Bloch sphere, Q-Sphere multi-qubit phase discs, and measurement histograms.
- Instructor Dashboard**
Failure Topology Heatmap with 1-click remedial lab dispatch to struggling cohorts.

Tier 2: Core Engine & Security Services

Python 3.12 • FastAPI Modular Monolith • Pydantic v2 • In-Process Quantum Kernels

Security & API Gateway

- AST Security Sandbox:** Python AST closed-grammar parser; 24 injection vectors rejected.
- Model Validator:** Canonical Pydantic Circuit Model JSON.
- Export Engine:** OpenQASM 3.0 compiler & code generator.

Multi-Engine Quantum Runtime

- Qiskit Aer 0.17:** Pre-warmed statevector & density matrix simulator (<150ms latency).
- PennyLane 0.45:** default.qubit mathematical conformance adapter.
- Google Cirq:** Endianness Rosetta Stone normalizer.
- State Trace Engine:** Causal tensor generation per gate.

Evidence-Bound AI Pedagogy

- Flight Recorder Engine:** 40+ Misconception Decision Matrix.
- Claim Validator:** Intercepts hallucinations; 100% mathematical probability consistency.
- Socratic Counterexample:** Minimal failing circuit synthesis.
- Curated Fallback:** 100% offline pedagogical reserve.

**TIER 3: PERSISTENCE & SOVEREIGN DPI TIER**

MeitY Empaneled Cloud • National Knowledge Network • DigiLocker / NAD

MongoDB Atlas M30
Async PyMongo driver with 26 collection indexes for sub-10ms query latency.

In-Memory Seed Store
Deterministic offline fallback (DEMO_LOCAL=1) for zero-internet execution.

NIC MeghRaj & NKN
Sovereign cloud hosting and high-speed campus transit across 1,500+ universities.

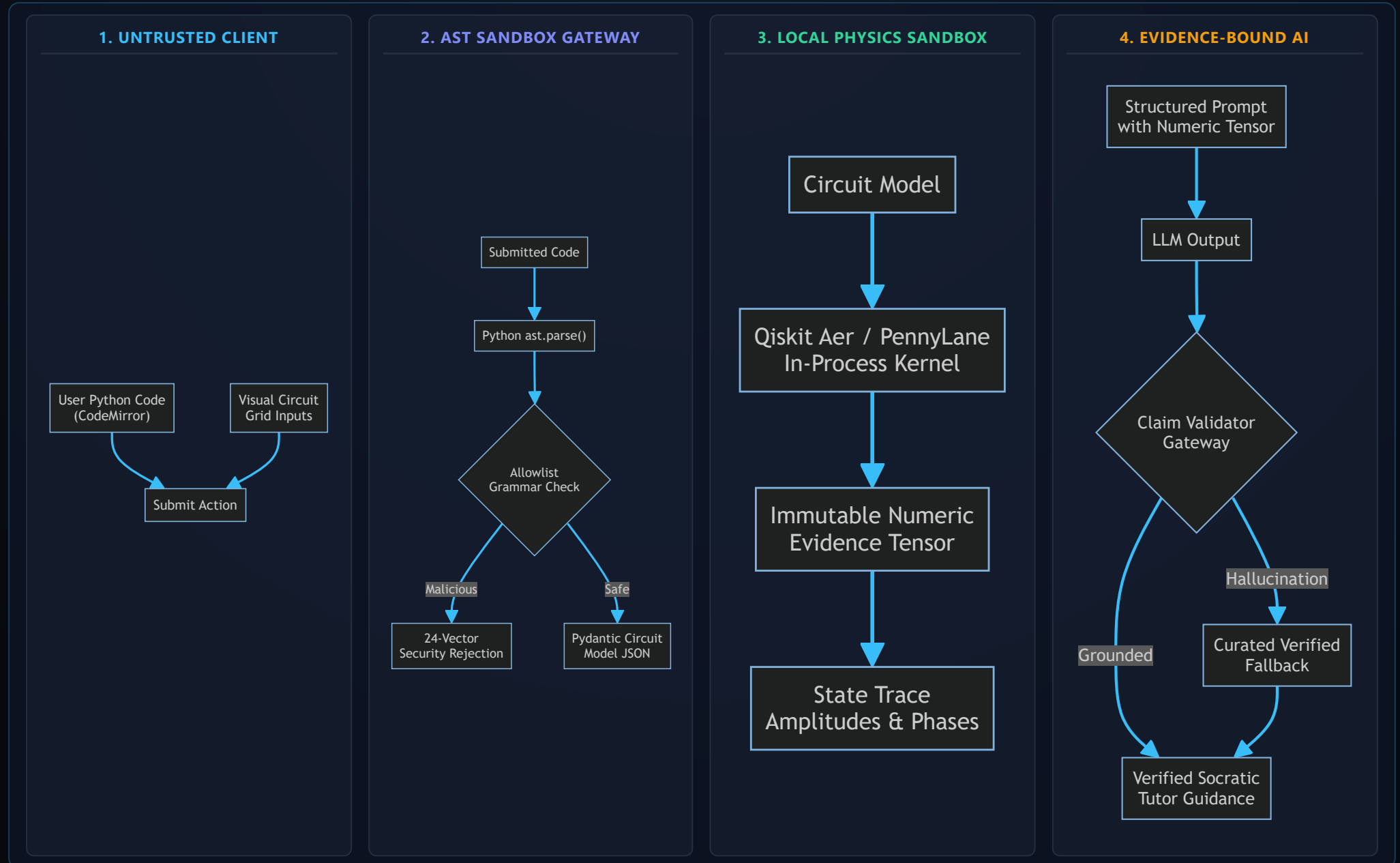
NAD & DigiLocker Sync
Cryptographically signed, tamper-evident (SHA-256) quantum micro-credentials.

2. Safety & Trust Boundary Architecture

SECURITY

ZERO HALLUCINATION

Dual Sandbox Strategy: AST Compiler Security for Code Execution & Mathematical Claim Validation for AI Explanations

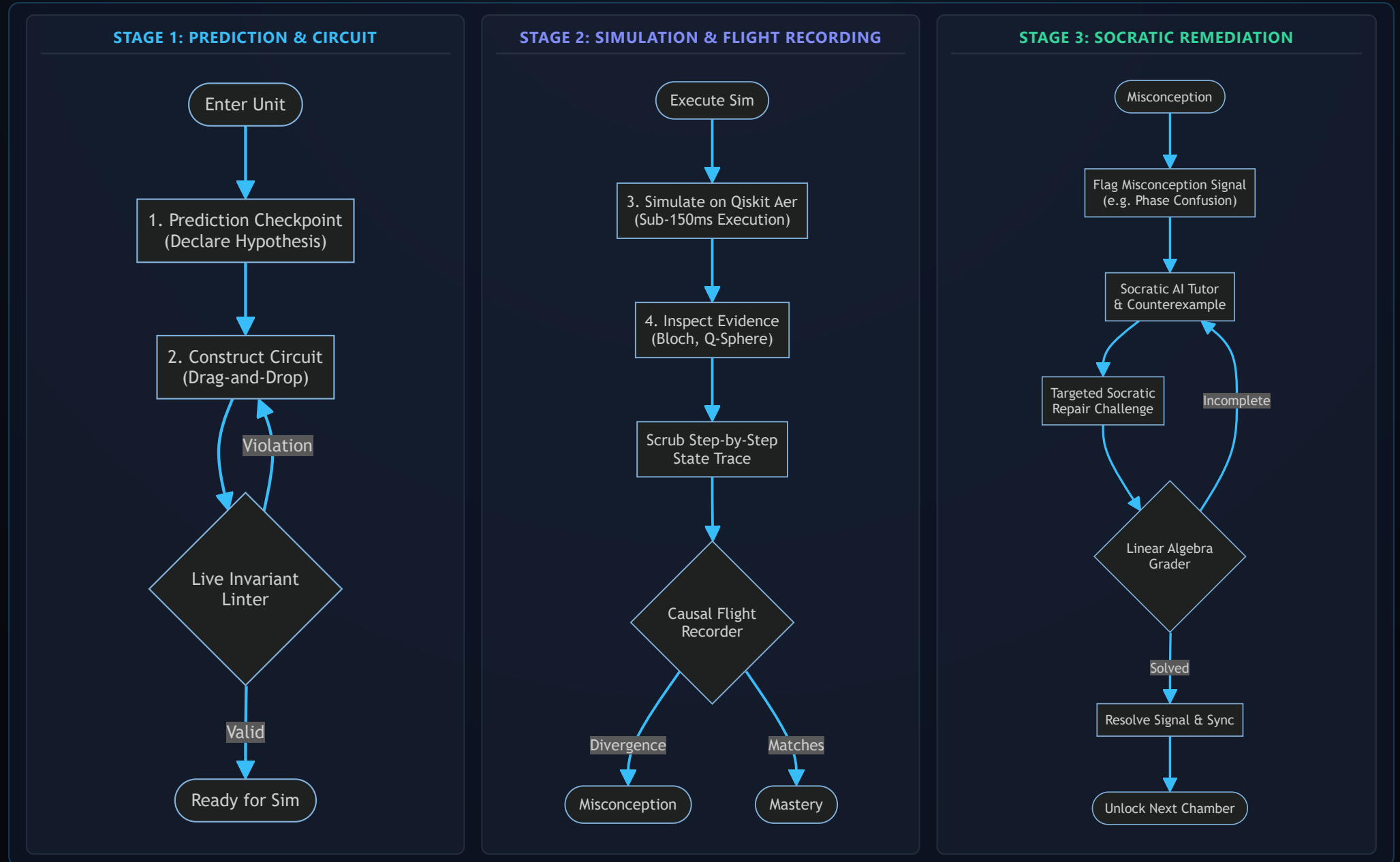


3. Core Learner User Flow: "Predict → Simulate → Inspect → Repair"

LEARNER UX

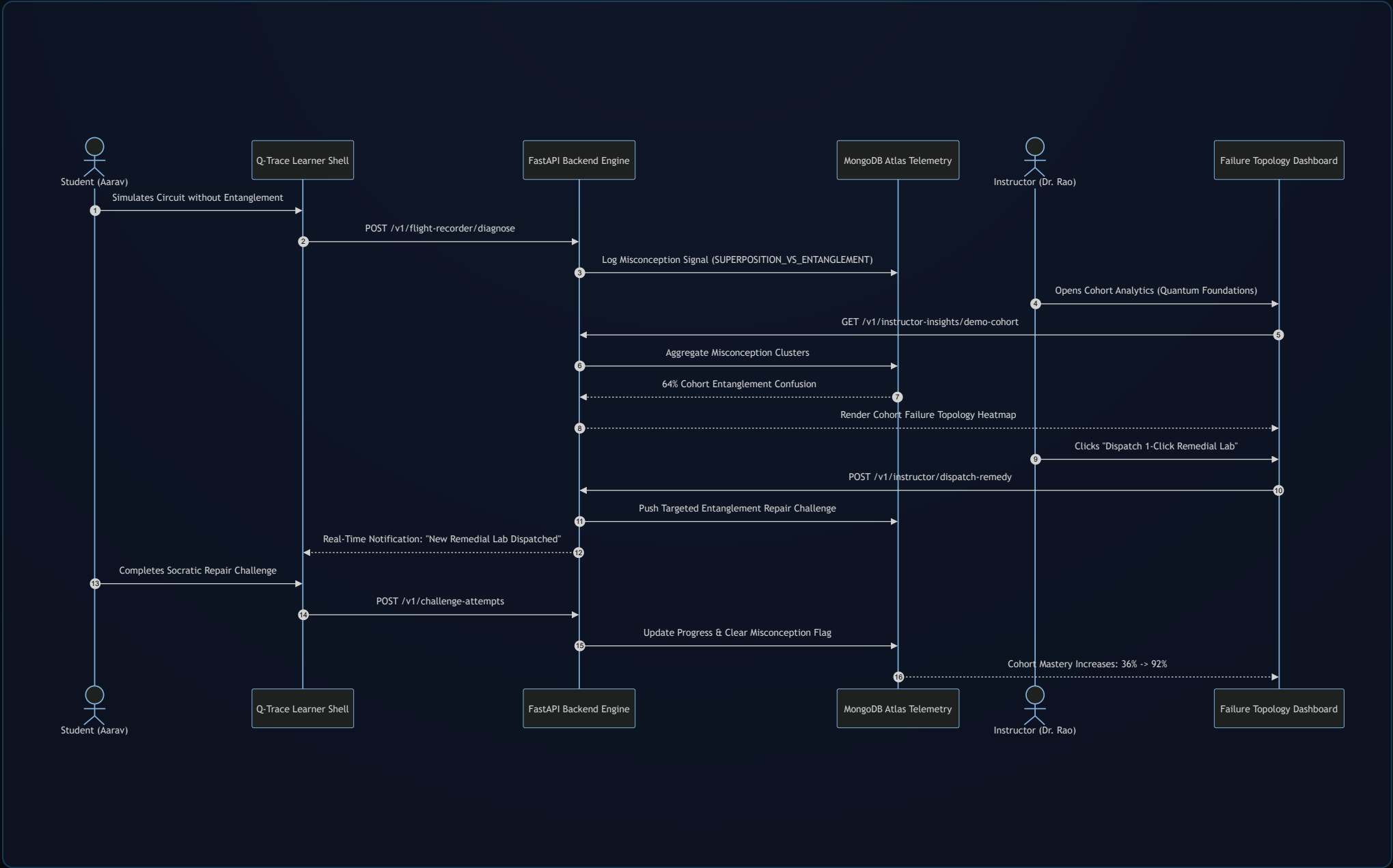
PEDAGOGY

Pedagogical Discovery Loop: Causal Misconception Diagnosis, Socratic Guidance, and Target Invariant Repair



4. Instructor Cohort Analytics & 1-Click Remediation Flow

Closed-Loop Institutional Governance: Telemetry Ingestion, Failure Topology Heatmap, and Instant Targeted Dispatch



5. National Credentialing & DigiLocker Issuance User Flow

DIGILOCKER

NQM CERTIFIED

Verifiable Human Capital Development: Proctored Capstone Assessment, SHA-256 Digest, and Sovereign DPI Credentialing

